

Survey on Fake Profile Detection on Social Sites by Using Machine Learning Algorithm

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Abstract— To avoid the spam message, malicious and cyber bullies activities which are mostly done by the fake profile. These activities challenge the privacy policies of the social network communities. These fake profiles are responsible for spread false information on social communities. To identify the fake profile, duplicate, spam and bots account there is much research work done in this area. By using a machine-learning algorithm, most of the fake accounts detected successfully. This paper represents the review of Fake Profile Detection on Social Site by Using Machine Learning.

Keywords— Fake Profile, Bots Accounts, Spam Profile, Machine Learning, Supervised Machine Learning, Support Vector Machine(SVM), Security Issues, Naïve byes, Decision Tree, Artificial Neural Networks(ANNs).

I. INTRODUCTION

We are living in an era where social networks become primary needs. Online social sites attracted millions of users from the last 20 years. Nowadays having accounts on social site is the priorities for the teenager. Social media platforms help to make new friends all over the world. It also helps to transfer video, audio, message and enhance the skills, ability to learn new things. Business organizers and government organizers also take advantage of social sites to provide their services in a better manner. These platforms make lifestyle so fast and smart. Online social media platforms have many broad applications. However, they are also having a dark side, in this having many malicious activities and we see that lots of malicious activity done by fake accounts. The fake accounts challenge the security of the system. To avoid all these problems there is a lot of research work done. Most of the research work completed in the field of supervised machine classifiers and unsupervised machine classifiers of the machine learning algorithm. Supervised machine learning algorithm uses classifiers such as naïve bays, decision tree, SVM, ANNs and also uses NLP(natural language process).through all these classifiers detect the fake accounts on social media. For the detection of fake accounts, the first process is to select the target profile for analysis to extract the feature set of the profile, such as name, chat history, location, friends list, followers, likes, comments, and tagging. Then applied the supervised or unsupervised machine learning classifier on it, which determines the target profile is fake or genuine. On this technique, there is a lot of work done and gives a successful detection result. Most of the machine learning algorithm helps in enhancing the accuracy rate of the system that is becoming 50%-96%.

II. LITERATURE SURVEY

In this paper Nambouri Sravya, Chavana Sai praneetha, S. Saraswathi [1] has proposed a system for identifying malicious activity on social sites. Nowadays social sites are a serious threat on the internet. To detect fake profile there are many models are proposed. Here, they focused on the Sybil and troll identities using Machine Learning Algorithm. Supervised Machine Learning Algorithm is recycled to overcome the problem. Sybil and troll accounts use an advanced technique, the data sets are collected by large data blogs then stored, if data is initiated malicious then data is clean and store again, after which cleaned shows the cleaned fake individualities and missing areas are fake individualities. Before clean store process, data is stored in a non-relational database for future reference and helps to remove the fake profile.

M. Smruthi, N. Harini [2] has used a hybrid model and skin detection algorithm to detect fake accounts on social sites. The strength of the proposed work is to detect the fake account with high accuracy. The result of this proposed work is using 400 mixed five Supervised Machine Learning Algorithm dataset of fake and real accounts. Here there 200 fake, 200 real accounts calculate the accuracy of the Supervised Machine Learning Algorithm. On the other hand, the skin detection algorithm is also used. Based on the skin-exposed percentage, the image is collect for the fake and real accounts calculated. 10-fold cross-validation was applied data set by using a supervised machine- learning algorithm. Dataset uses 13 features of user identity. Here supervised machine learning algorithm uses a decision tree classifier having the highest accuracy that is 80% rest of the classifier gave 60% to 80% accuracy with a 20% error rate. The accuracy rate of the KNN algorithm 60% and other classifiers helps to increase the accuracy rate up to 80% such as Naïve Bayes and Decision tree classifier. The skin detection algorithm first was implanted deep learning algorithm for the detection of an image which image is human or not. If the image is a human being, then apply the skin detection algorithm and calculate the skin exposed percentage. If an image containing more than 13 percentage skin exposed percentage, it could be consider as a fake profile because most of the fake profile is taken a high percentage of skin exposed area or not having any image. If the skin detection percentage of the image is high, accounts are considered as fake accounts.

Pooja Tehlan, Rosy Madaan, and Komal Kumar Bhatia [3] have proposed work to detect spam by expending Machine Learning Algorithm, Fuzzy logic, and Artificial intelligence.

This approach overcomes limitation spamming by using the Semi- Supervised Learning algorithm. Hereby using fuzzy logic handle the large data set very efficiently and detect the spam within a second, in this paper. it reduces the cost efficiency, time and use of complex software. The fake accounts on the social site are responsible for spamming. The spamming could access the personal information and start back mail them. According to some reports, spam reports spam contains the account URL. Even though.

filtering Facebook calculated 87% i.e. 955 million accounts are fakes. Spam is responsible for wasting the time of users and consumes network bandwidth. The fuzzy logic model is developing to filter the large data set efficiently and detect the spam without any pre-tanning. It also proposed a multi- layer perceptron for the analysis of work. This technique helps to gain a more accurate and fast result as compare to another technique.

P. Srinivas Rao, Dr. Jayadev Gyani and Dr.G.Narsimha [4] approach Machine Learning and NLP method to increase the accuracy rate of fake accounts detection. Through support vector and Naïve bays, many problems are removed on social media, the problem is like privacy issues, cyber bullies, and troll, etc. Most of the time seen these types of malicious activities are done by false accounts present on social sites. This paper takes the Facebook dataset for the recognition of false accounts. Here Machine Learning Algorithm along with NLP pre-processing method applies for analyzing the dataset and for classification of the profile use support vector machine classifier that helps to detect the profile is fake or genuine.

Rohit Raturi [5] is focused on the architecture of the system. The approach two concepts of architecture in the first architecture will identify the accounts detail by using NPL and network identifiers. Based on network identifier if two or more than two accounts are accusing by users they need to ask them to provide security. In second architecture uses the Support Vector Machine BOW (Bag of Words) concept for the identification of the number of the words. The number of words is harmful, the harmful words collected as the dataset. BOW with predication mode with SVM (Support Vector Machine) attach the dataset and divide the dataset for testing and training. It helps to calculate harmful words from the individual accounts and identify the false account based on the content present in the accounts then send a warning message for providing authentic prove to continue with the account. If the scale is greater than three, the user must provide proof to continue with the account. The main problem of a social network is not providing authentication before publishing the data. In this paper, they used Facebook and Twitter datasets such as twitter tweets, tag, likes, and posts.

They use sentiment analysis to identifying the goals. For better implementation of NLP and SVM, the text classifier, which is considered as a parent algorithm for the text classifier, helps to save space data of social networks. Both CNB and BOW work on the weka platform. Weka is a data test tool that provides the mode and enhances the accuracy percentage rate.

Estee Vander Walt and Jan Eloff [6] are discusses on an engineered feature such as a friend list and the number of follower order of fake profiles for successful detection of

the false profile is created by a human on the social media platform. Here, for the identification of social media accounts, there are many attributes are used such as name, location, and profile image. Human and bots accounts have similar characteristics such as human accounts have a name then bots generated same accounts through an engineered feature. In this paper, determine the three machine learning modules, which are used for successful recognition of fake profiles. For identification of engineered features that which one of the accounts performed well or not is done by entropy present there. As a result, FI score 49.7% by chance and alone it could predict 50% which is not optimal. Here the machine learning model (MLA) is used to train the data without any dependency on the behavioral data set. Sometimes MLA is missing the viral bluff in the large data set.

Sumit Milind Kulkarni and Prof. Vidya Dhamdhare [7] proposed a model, which is based on programming structure through the programmed structure to identify the online profile, is fake or genuine. This programmed model has used some classifier techniques such as SVM, Naïve bays and Decision Tree for identification of profile. Initially select the target profile, which is going to be tested, then select the suitable feature i.e. feature where classification algorithm implemented. The feature extracted to the trained classifier, a classifier trained the data with the new dataset. The classifier is determining the target profile is real or fake, if the account is fake, sends a notification to that profile for sending an authentic identity. Valid identification submitted once more is sending a feedback message to a classifier that the profile is not fake. By using NLP it designs a framework to detect the fake profile with high efficiency. NLP improves the accuracy rate of up to 95% for the detection of any online social sites profile. This method is detecting the fake profile in fewer efforts. Ala' M. Al-Zoubi, Ja'far Alqatawna, and Hossam Faris [8] have developed a model for spam profile detection. For this detection, they took 82 twitter's profile dataset available feature sets. The classifier of spam profile they can use 10 binary and simple feature. For feature selection, there are two methods are used first one is relief F and second is information gain. The classification algorithm is used such as Naïve bays, k-Nearest Neighbor, Multilayer Perceptron and Decision Tree. The feature set is taken out of presented profile data unconcerned linguistic used by the users' tweets seen that the promising result which is obtained by the use of decision tree and naïve byes classifier. The result reveals the suspicious word and repeated word with high influence and truthfulness of the detection process unconcerned of linguistic used for

tweets by users.

Yuval Elovici, Moshav Arugot (IL), Michael Fire, Netanya (IL); Gilad Katz, Rehovot (IL) [9] are inventing a method for protecting the privacy of users on online social networks. In this invention, they took the fake and legitimate profile data of the existing social site and then feature is extracted from the pre-determined set of the target fake and legitimate profile, the extract feature set chosen from the friend's list and followers of the target profiles. If examining the relationship of each private node and between the social accounts. Classifiers are applied to other existing features of fake profile, by using trained and non-structured supervised machine learning.

Suheel Yousuf Wani, Mudasir M Kirmani and Syed Imamul Ansarulla [10] are explaining a theoretical model

for the recognition of fake accounts. The proposed work uses the machine learning algorithm such as SVM, Naïve Bays, Decision Tree, and ANNs. For the detection of fake profile applied supervised machine learning algorithm, sophisticated noise elimination and the data normalization technique data set. The artificial bee colony and ant colony optimization algorithm nature inspire are used to identify the non-significant attribute the dataset and do attribute reduction. For both fake and genuine profiles, the individual support vector machine algorithm was trained. Here the ensemble classifier is used to generate a more accurate prediction. The ensemble classifiers using the weka tool for the detection of fake profiles on Facebook. In this, the combination of more than two Machine Learning Algorithm is used the detection of false profiles and original profiles on Facebook.

This model was profitably applied by Wu et al.[15], Ahmazadeh and Abadi[16] and Gu et al.[17]. In their research work, they showed detect the bots account using the clustering method. The clustering technique is a common method in the Unsupervised Machine Learning Model. In this model, detect the bots by using data. Take the unlabeled data or group based data [19]. The bots accounts share similar characteristics and have the same purpose and fake human accounts not share the same purpose.

Venkatesan et al. [18] work on the reinforcement concept in this they prove that the model successfully detects the bots accounts itself. This model required a feedback environment that helps to improve the detection process of fake accounts corrects them effectively but the reinforcement model is just not available on social media. Arif et al. [20] through emerged features and a set of iterative rules detect spam on social media.

In filtering technique when a new malicious activity is identified then the sender will be a supplement to a black list. The approach of combat spam has been suggested to black list malicious URL body content on twitter and detention known as bots [21]. Sometimes the Spam filtering is not so easy, mislead the projected methodology, the spammer approaches the dynamically modified and automated technique. Humans shield themselves to escape easy detection. In the case of black listing, they simply create a new profile with fake identity [22] as soon as the detection is performed on the profile; the profile is going black listed.

III. DETECTION OF FAKE PROFILES

Humans or bots generate finding the fake profiles on the social network. First, create a user generated Machine Learning Module using fake profiles, through these algorithms easily find out which profile is fake.

- Initially clear whole the earlier data that are stood for the detection of fake profiles generated by humans or bots. We want to detect those profiles, which are generated by humans.
- From the research studies, we found that most of the real and fake accounts having a profile picture and name attribute on it.
- We also find that the profiles/accounts are genuine to have an average of more than 30 followers. Therefore, we should discard that profiles which have more than 30 followers.

- For identification of fake accounts first, we know that what is fake account is actually? So initially create at least ten or more than ten thousand fake profiles for having enough data which has more data than the algorithm may recognize the profile created by humans not by bots and also recognize what is the fake profile?[13]
- We found that after research studies in psychology, many users lied about their name, location, age, and gender. The create account on a social network with false identities and we are also found that who gave wrong information about their identities, are age set is 18-19 years.[10]
- We also see that the profiles are created on the social network are linked with email ids. So check email ids linked with the profiles, it is also a way to identify that the profiles are fake or genuine. In this, we check them name present in the email ids.
- We should also check the location as users set some impalpable places like the Pacific Ocean to stay safe but the accounts are retrieving from India.

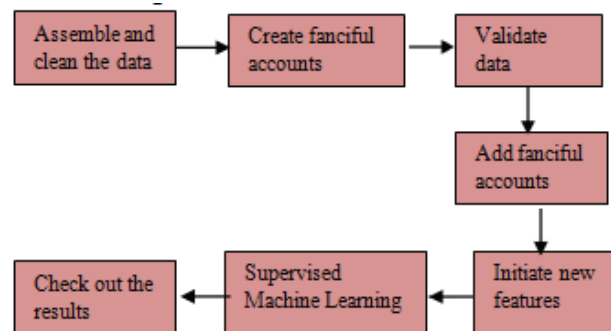


Fig. 1. Flow Diagram of to Detect the Fake Accounts

[11] The research work have been done to detect forty-botched accounts and to be eliminated and the cyborg cannot be used to differentiate the fake account created by humans. In recent times, the machine has developed as learning. We can easily separate fake accounts by implementing a data set with fake accounts, mark them, and act as real as possible. In this paper machine leakage, outline which provides a mixture of metadata, network and temporal traits to explore the extremist users and adopts content in social media and predicting mutual reciprocity. Here, unique datasets containing millions of two hundreds of users, and manually identified, informed and suspended on twitter because of joining extremist campaigns. Also, in this take advantage of millions of tweets generated by random samples of 25 thousand regular users; those were in touch or with the extremist material. They do three forecasts (I) to detect extremist users,

(II) to estimate whether regular users will adopt extremist materials, and finally (III) to estimate that the users will be able to get contacts started by extremists reed or not. All forecasts had been established in two places, post hoc (time independent) on total data, and the work of a real time prediction. The perfection of our framework is extremely promising in various forecast destinations from 93% AUC for extremist user detection, 80% AUC for content adoption prediction, and finally up to 72% AUC for interaction reciprocity forecasting. Now this concludes that by providing

feature analysis which helps to determine, initial signals that provide predictive power in different situations.

In this Learning Ebrahimi et al. [23] compared the Support Vector Machine Model to the Naïve Bayes Model where it represents a one-class Support Vector Machine exceed the binary classification model. In this norm, train the one-class Support Vector Machine with an outnumbered class [24]. In the social media site, it cannot practically possible to minimize an outnumbered class exist fake accounts [25]. As we know that in the Semi-supervised machine learning algorithm, there should be a fine line between the classes [26].

[14] In this approached a methodology is used which removes the sophisticated noise and data normalization technique on the dataset before analyzing them. This technique has been applied to detect the non-significant attributes in datasets and to do attribute discount accordingly by applying natural enthused algorithms like Artificial Bee Colony, Ant Colony Optimization. In this, paper the trained model using supervised machine algorithms independently for both data sets such as fake and genuine. Ensemble classifiers have been used for the prediction more accurately as shown in figure 2. Based on research analysis, there is no such model in being used for the detection of both fake and genuine profiles on social sites. Therefore, there are more than one machine learning algorithms are used for the detection of fake and genuine profiles.

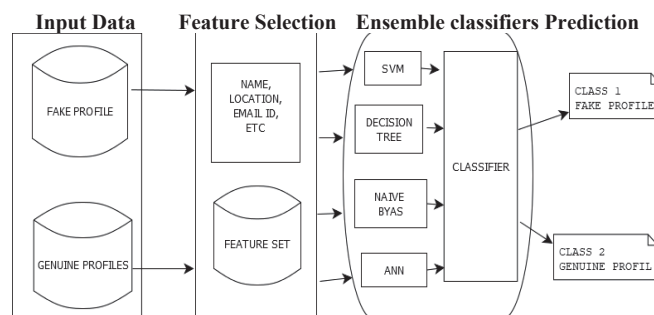


Fig. 2. Architecture of fake deduction

TABLE I. THE RESULT OF SUPERVISED MACHINE LEARNING [6]

Model	Accuracy	Fi-score	PR-AUC
SVM linear	68.05%	32.16%	27.76%
Reinforcement	87.11%	49.75%	49.90%
Adaboost	85.91%	47.54%	49.53%

IV. CONCLUSIONS

Now in the end, we conclude that research works are to be done to identify or detect the fake profile created by false users. The bots cannot used to make a distinction that fake profile generates by human beings. The Machine Learning Module is now progressing in the present day. We can discriminate a fake profile without difficulty by using data sets with fake profiles then detect them as fake profiles or real profiles. Therefore, after that the model will successfully determine the profile is create by a human is real or fake profiles when they truly existing data set will be give out. There are many machine learning modules that are used to detect both genuine and fake profiles on social sites.

For the detection of fake profiles, both supervised and unsupervised machine learning is used. These modules are successfully detecting fake accounts.

REFERENCES

- [1] Nambouri Sravya, Chavana Sai praneetha, S. Saraswathi, "Identify the Human or Bots Twitter Data using Machine Learning Algorithms", International Research Journal of Engineering and Technology (IRJET), Volume: 06 Issue: 03 | Mar 2019 www.irjet.net, e-ISSN: 2395-0056, p- ISSN: 2395-0072.
- [2] M. Smruthi, N. Harini, "A Hybrid Scheme for Detecting Fake Accounts in Facebook", International Journal of Recent Technology and Engineering (IJRTE) ISSN: 2277-3878, Volume-7, Issue-5S3, February 2019.
- [3] Tehlan, Pooja, Rosy Madaan, and Komal Kumar Bhatia. "A Spam Detection Mechanism in Social Media using Soft Computing."
- [4] Rao, P. S., J. Gyani, and G. Narsimha. "Fake profiles identification in online social networks using machine learning and NLP." Int. J. Appl. Eng. Res 13.6 (2018): 973-4562.
- [5] Raturi, Rohit. "Machine learning implementation for identifying fake accounts in social network." International Journal of Pure and Applied Mathematics 118.20 (2018): 4785-4797. J. Wang, "Fundamentals of erbium-doped fibre amplifiers arrays (Periodical style—Submitted for publication)," IEEE J. Quantum Electron., submitted for publication.
- [6] Van Der Walt, Estée, and Jan Eloff. "Using machine learning to detect fake identities: bots vs humans." IEEE Access 6 (2018): 6540-6549.
- [7] Kulkarni, Sumit Milind, and Vidya Dhamdhare. "Automatic detection of fake profiles in online social networks." Open access international journal of science and engineering 3.1 (2018): 70-73. M. Young, The Technical Writers Handbook. Mill Valley, CA: University Science, 1989.
- [8] Ala'M, Al-Zoubi, Ja'far Alqatawna, and Hossam Faris. "Spam profile detection in social networks based on public features." 2017 8th International Conference on information and Communication Systems (ICICS). IEEE, 2017.
- [9] Elovici, Yuval, and Gilad Katz. "Method for detecting spammers and fake profiles in social networks." U.S. Patent No. 9,659,185. 23 May 2017.
- [10] Gurajala, Supraja, et al. "Profile characteristics of fake Twitter accounts." Big Data & Society 3.2 (2016): 2053951716674236.
- [11] Ferrara, Emilio, et al. "Predicting online extremism, content adopters, and interaction reciprocity." International conference on social informatics. Springer, Cham, 2016.
- [12] Caspi, Avner, and Paul Gorsky. "Online deception: Prevalence, motivation, and emotion." Cyber Psychology & Behaviour 9.1 (2006): 54-59.
- [13] Bergen, Emilia, et al. "The effects of using identity deception and suggesting secrecy on the outcomes of adult-adult and adult-child or-adolescent online sexual interactions." Victims & Offenders 9.3 (2014): 276-298.
- [14] Wani, Suheel Yousuf, Mudasir M. Kirmani, and Syed Imamul Ansarulla. "Prediction of fake profiles on Facebook using supervised machine learning techniques-A theoretical model." International Journal of Computer Science and Information Technologies (IJCSIT) 7, no. 4 (2016): 1735-1738.
- [15] Wu, W., Alvarez, J., Liu, C. and Sun, H.M., 2018. Bot detection using unsupervised machine learning. Microsystem Technologies, 24(1), pp.209-217.
- [16] M. Yahyazadeh and M. Abadi, "BotOnus: An online unsupervised method for botnet detection," ISC Int. J. Inf. Secur., vol. 4, no. 1, pp. 51_62,2012.
- [17] G. Gu, R. Perdisci, J. Zhang, and W. Lee, "BotMiner: Clustering analysis of network traf c for protocol-and structure-independent botnet detection," in Proc. USENIX Secure. Symp., vol. 5. 2008, pp. 139_154.
- [18] S. Venkatesan, M. Albanese, A. Shah, R. Ganesan, and S. Jajodia, "Detecting stealthy botnets in a resource-constrained environment using reinforcement learning," in Proc. Workshop Moving Target Defence, 2017, pp. 75_85.
- [19] P. Galán-García, J. G. De La Puerta, C. L. Gómez, I. Santos, and P. G. Bringas, "Supervised machine learning for the detection of troll problems in Twitter social network: Application to a real case of cyberbullying," Logic J. IGPL. vol. 24, no. 1, pp. 42_53, 2015.

- [20] M. H. Arif, J. Li, M. Iqbal, and K. Liu, "Sentiment analysis and spam detection in short informal text using learning classifier systems," in *Soft Computing*. Berlin, Germany: Springer, 2017, pp. 1_11.
- [21] K. Thomas, C. Grier, D. Song, and V. Paxson, "Suspended accounts in retrospect: An analysis of Twitter spam," in *Proc. ACM SIGCOMM Conf. Internet Meas. Conf.*, 2011, pp. 243_258
- [22] S. Cresci, R. Di Pietro, M. Petrocchi, A. Spognardi, and M. Tesconi, "Fame for sale: Efficient detection of fake Twitter followers," *Decision Support Syst.*, vol. 80, pp. 56_71, Dec. 2015.
- [23] M. Ebrahimi, C. Y. Suen, O. Ormandjieva, and A. Krzyzak, "Recognizing predatory chat documents using semi-supervised anomaly detection," *Electron. Imag.*, vol. 2016, no. 17, pp. 1_9, 2016.
- [24] C. Bellinger, S. Sharma, and N. Japkowicz, "One-class versus binary classification: Which and when?" in *Proc. 11th Int. Conf. Mach. Learn. Appl. (ICMLA)*, vol. 2, Dec. 2012, pp. 102_106.
- [25] L. M. Jupe, A. Vrij, G. Nahari, S. Leal, and S. A. Mann, "The lies we live: Using the variability approach to detect lying about occupation," *J. Articles Support Null Hypothesis*, vol. 13, no. 1, pp. 1_13, 2016.
- [26] C. Beleites, K. Geiger, M. Kirsch, S. B. Sobottka, G. Schackert, and R. Salzer, "Raman spectroscopic grading of astrocytoma tissues: Using soft reference information," *Anal. Bioanal. Chem.*, vol. 400, no. 9, p. 2801, 2011.